Module 9 Lecture - Multivariate Correlation

Research Methods in Psychology

Quinton Quagliano, M.S., C.S.P

Department of Psychology

Table of Contents

1 Objectives and Overview 2
- 1.1 Learning Objective(s) 2
- 1.2 Chapter Overview 2

2 Review of the Casual Criteria 3
- 2.1 Overview 3

3 Longitudinal Designs and Temporal Precedence 3
- 3.1 Overview 3
- 3.2 Cross-sectional Correlation 4
- 3.3 Autocorrelations 4
- 3.4 Cross-lag Correlation 5
- 3.5 Longitudinal and Causation Criteria 5
- 3.6 Why Not Experiment 6

4 Multiple Regression and Third Variables 7
- 4.1 Overview 7
- 4.2 Criterion and Predictor Variables 7
- 4.3 More Than Two Variables 7
- 4.4 Statistics for Third Variables 8
- 4.5 Beta coefficients 8
- 4.6 Interpretation Example 9
- 4.7 More Predictors, The Merrier? 9
- 4.8 In Popular Media 10
- 4.9 Still Not Causal 10

5 Pattern and Parsimony 10
- 5.1 Overview 10
- 5.2 Meaning 10
- 5.3 In Popular Media 11

6 Mediation 11
- 6.1 Overview 11
- 6.2 Compared to “Third Variable” 12
- 6.3 Compared to “Moderators” 12

7 Analysis with the Four Validities 13
- 7.1 Overview 13
  - 7.1.1 External Validity 13
  - 7.1.2 Statistical Validity 13
  - 7.1.3 Construct Validity 14

1 Objectives and Overview

1.1 Learning Objective(s)

- State why simple bivariate correlations are not sufficient for establishing causation.
- Explain how longitudinal correlational designs can establish temporal precedence.
- Explain how multiple-regression analyses can rule out some (but not all) third variables.
- Describe the value of pattern and parsimony, in which a variety of research results support a single, parsimonious causal theory.
- Explain the function of a mediating variable.
- Understand the relative strengths and weaknesses of a multivariate design, compared to bivariate designs
- Be able to describe interpreting longitudinal and multiple regression designs, and interpret statistical values from these designs

1.2 Chapter Overview

- _____ $\rightarrow$ multiple (more than two) variables
  - We will still be using a lot of correlation-like techniques
- Example: In a _____ design, I might have just depression and anxiety that I am looking at (two variables). In a multivariate version, may I will have depression, anxiety, well-being, and academic success.
- This correlational research is still comprised purely of _____ variables
  - We still don't have manipulation, yet
- The techniques and designs discussed today will get us closer to a causal claim, but we are still just short of reaching the necessary criteria, and still in range of _____ claim

2 Review of the Casual Criteria

2.1 Overview

- Covariance: are _____ (or more) variables varying in some consistent pattern?
- Temporal Precedence: does one variable come _____ another in time?
- _____ Validity: is the relationship between our variables not better explained by a third variable?

? Which of these is already met by a bivariate design?

- ◉ A) Covariance
- ○ B) Temporal precedence
- ○ C) Internal validity
- ○ D) None of them

Explanation:

3 Longitudinal Designs and Temporal Precedence

3.1 Overview

- Longitudinal Research:
  - Research conducted over multiple _____ points (usually over an extended period of time)
  - Normally follows the _____ group of people taking same multiple measures at the many time points
- Example: I study people's rating of their own physical and mental health health over 5 years, taking measurements every half-year.
- The "multivariate" nature of this comes from the 2 or more measurements happening at _____ time points

  - In essence, the same measures are treated as _____ variables because they fall at different time points

- With this design, comes multiples types of _____, we will discuss each type separately in a moment

3.2 Cross-sectional Correlation

- Are the variables at the same time point correlating with one _____?
- The _____ refer to the separate measures

? Which of the following r values suggest at least weak relationship?

- (●) A) r = .007
- ( ) B) r = .070
- ( ) C) r = .138
- ( ) D) r = .099

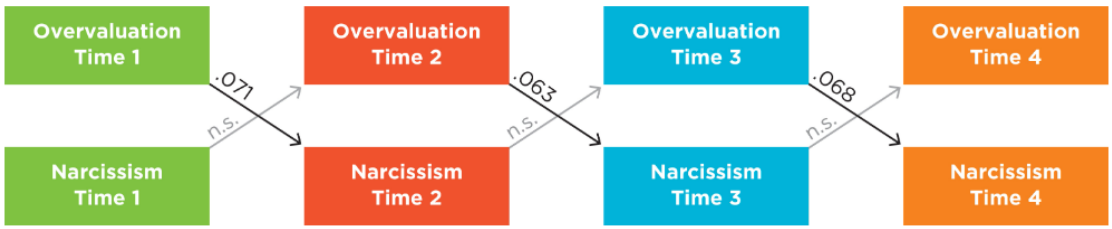
Explanation:

3.3 Autocorrelations

- Are the variables correlated with _____ at the other time points?

- "Auto-" as a root prefix means "self", so this is a measure correlated with itself

3.4 Cross-lag Correlation



- Are the variables correlated with each other _____ time points?
- This is sufficient for establishing _____
- This is also the _____ outcome we are looking at for longitudinal designs
- The "cross" here should remind you of the diagonal direction of the arrow in this diagram

Discuss: Normally these are interval/ratio variables, but what statistic could we use if they are ordinal?

3.5 Longitudinal and Causation Criteria

What is the one causal criteria we are still missing in a longitudinal design?

- (●) A) Internal validity
- ( ) B) Covariance
- ( ) C) Temporal precedence
- ( ) D) All of them are met

Explanation:

- However, there needs to be an _____ effort in order to rule out third variables
  - This can be done by including _____ variables to track which may have an impact on one or both of the variables
  - Example: If I am worried that energy level may be affecting the relationship between exercise amount and self-worth, I should measure that as well.
- But remember, that even if we get pretty good evidence for all the _____ criteria - we are not fully there yet.

3.6 Why Not Experiment

- You can't always easily _____ variables that may be causing something

Discuss: What are examples of variables that are difficult to manipulate for practical reasons?

- You can't always _____ manipulate a variable

Discuss: What are examples of variables that are difficult to manipulate for ethical reasons?

- Sometimes we can do a _____ study that would be unethical to over a longer period of time. Then we can combine those experimental designs with longitudinal designs.

  - Example: We can sleep deprive people for one night, but can't do it ethically for an entire month.

4 Multiple Regression and Third Variables

4.1 Overview

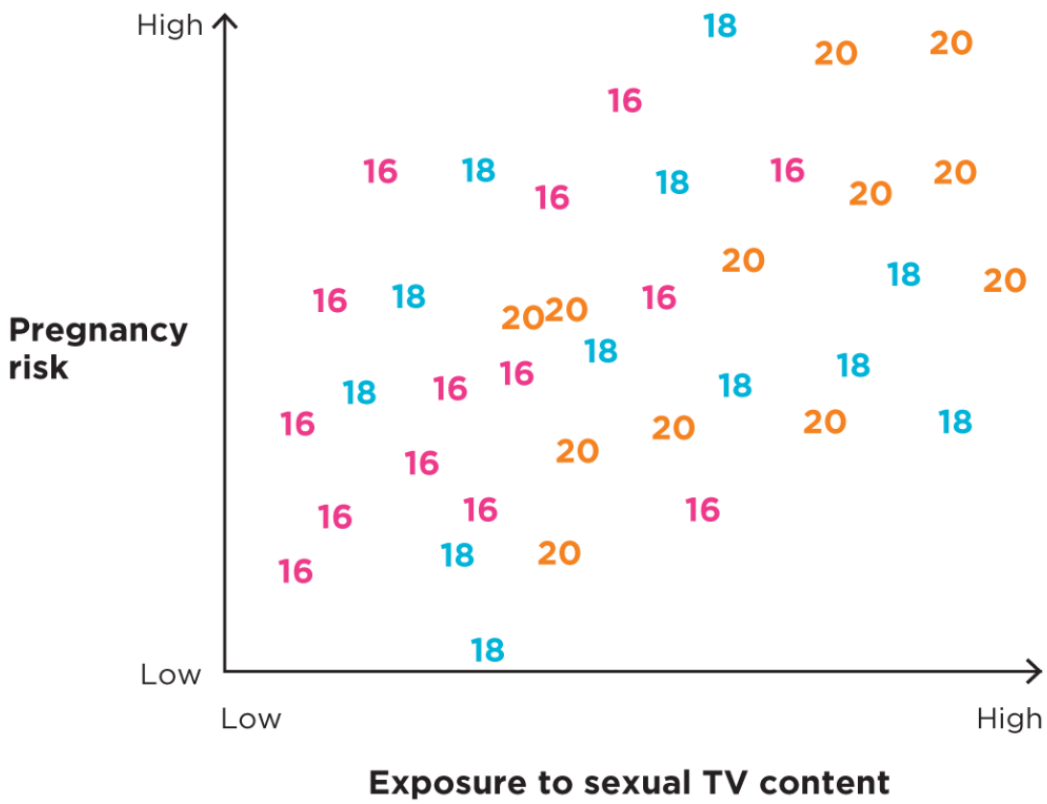
- Multiple _____ regression is when we predict one continuous variable by way of multiple _____ variables
  - Stats sidebar: some books temporarily refer to this as "multivariate regression" - don't call it that, because that is a separate, more complicated technique
- Technically, we can predict with _____ or continuous variables, but our single outcome should be continuous
  - Stats sidebar: outcomes of multiple regression can be binary/categorical, but this becomes much more complicated and won't be our focus in this class

4.2 Criterion and Predictor Variables

- When we use multiple regression, we have:
  - _____ the "dependent" variable we are trying to predict
  - Predictor variables: Those that we are using to _____ the criterion (duh)
- In summary: multiple predictor variables $\longrightarrow$ single criterion variable

4.3 More Than Two Variables

- The benefit to this is we can _____ control for the effects of other variables and _____ the effects of the variables we care about the most



4.4 Statistics for Third Variables

- When we talk about controlling for some variable in a model, we are talking about "holding it still" or _____ the effect it has on the criterion or _____
- That way, we can see the individual _____ of each predictor variable on the criterion
- This is not the same as making a variable a _____ in the design itself - this is purely a statistical procedure

4.5 Beta coefficients

- Beta, represented as β, is a _____ effect that one unit on the predictor variable changes on the outcome variable
- For example, if our age β is +1.25 on pregnancy risk - that means that for every 1 year of age, _____ pregnancy risks raises by 1.25.

- A beta further from 0 signifies a _____ effect, and we can look for statistical _____ in β just like in other stats.

Discuss: If I have a beta of 4.00 on predictor X for criterion Y, provide an interpretation of the relationship between the two variables.

4.6 Interpretation Example

Multiple-Regression Results from a Study Predicting Pregnancy from Sexual Content on TV and Age

CRITERION (DEPENDENT) VARIABLE: PREGNANCY RISK	BETA	95% CI FOR BETA	Statistical Significance
Predictor (independent) variables:			
Exposure to sex on TV	0.25	[.14, .36]	*
Age	0.33	[.20, .46]	*

Note: Data are fabricated, based on imagined results if the researchers had used only two predictor variables.

*$p < .05$, meaning the result is statistically significant and the 95% CI does not include zero.

4.7 More Predictors, The Merrier?

- There is no natural limit to however many predictors to put into a model, with some caveats:
  - A good rule of thumb is that sample should be at least _____ times the number of predictor variables
  - Too many predictors can _____ effects if the predictors correlate with one another!

Discuss: AI is sometimes described as an especially complicated multiple regression model - why do you think this is?

4.8 In Popular Media

- "Controlling for", "Adjusting for", "Considering" are all phrases that often indicate use of a _____ model. Be on the lookout for these.

4.9 Still Not Causal

- Despite the fact that longitudinal and multiple regression studies are _____ at statistically control of third variable and establishing temporal precedence…
- They don't quite reach the _____ gold standard of experiments

5 Pattern and Parsimony

5.1 Overview

- Sometimes the _____ of evidence from substantial correlation studies appears to be sufficient for establishing causality… is it?

5.2 Meaning

- _____ is the principle that says we should seek the simplest possible (and still accurate) description of a phenomenon or relationship
- The "pattern" we are speaking of here is the pattern of _____ coming from numerous studies

? What type of source is going to give a nice quantitative overview of the weight of evidence in an area?

- (●) A) Original empirical journal article
- ( ) B) Meta-analysis journal article
- ( ) C) Literature review journal article
- ( ) D) Scientific journalism

Explanation:

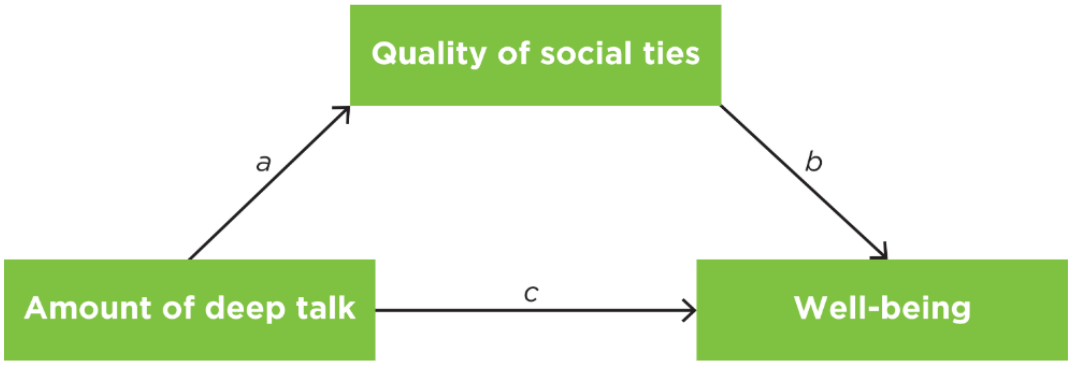
5.3 In Popular Media

- Popular media tends to like big _____ papers that seem to point out a prominent pattern in research
- However, remember the _____ associated with putting too much trust in scientific journalism

6 Mediation

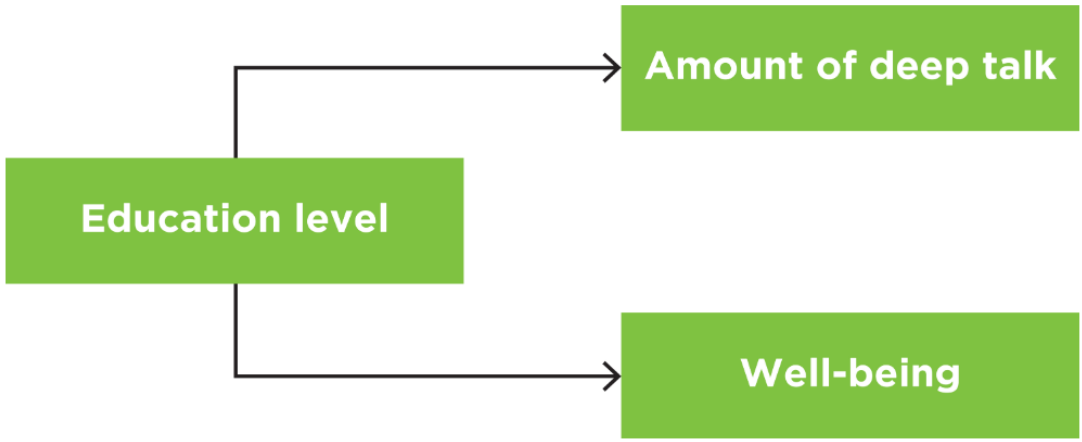
6.1 Overview

- Mediation: a claim that one variable's relationship with another is better _____ by another variable, i.e., some mechanism through which an effect occurs - the "why"
- Technically this is a _____ claim, but is often hinted at via correlational designs.



6.2 Compared to "Third Variable"

- A "third variable" more so refers to a variable unrelated to a mechanism that just _____ to be related to both variables of interest



▶ More Information

6.3 Compared to "Moderators"

- A _____ is best described as a variable that's state changes the relationship between two others
- E.g., The link between depression and anxiety are stronger when trauma is high

Discuss: How common do you suspect mediators and moderators are in psychological phenomena?

7 Analysis with the Four Validities

7.1 Overview

- How do we investigate these designs? Largely the same as the bivariate designs!

7.1.1 External Validity

- How was the sample _____? To a lesser extent, how many people are in the sample, and what are their demographics?

7.1.2 Statistical Validity

- What is our _____ (effect size)?
- What is our precision (_____ intervals)?
- Are our statistics _____?
- Do these results _____ in other studies?
- Do we have a restriction of _____ or outliers?

? Which of the following scenarios seems most indicative of poor precision?

- (●) A) We have an r-squared of 0.25
- ( ) B) Our p-value is 0.10
- ( ) C) Our effect isn't shown in other studies
- ( ) D) Our confidence intervals for r are [0.20, 0.80]

Explanation:

7.1.3 Construct Validity

- Are our measures consistent and _____?
- Are our measures accurate and _____ to our intended construct?

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms